CPE314 Data Communication and Computer Networks

Lecture 5: Transmission Impairment

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Lecture Agenda

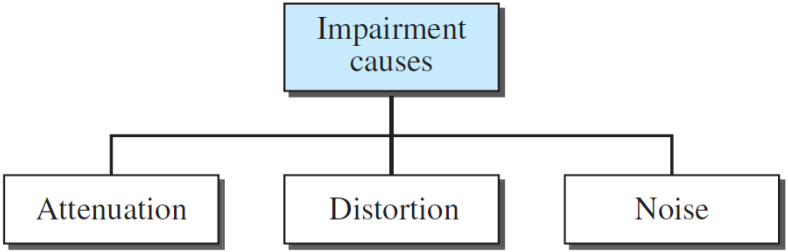
1. Transmission Impairment
2. Attenuation
3. Distortion
4. Noise
5. Summary

Transmission Impairment

Definition

Signals travel through imperfect transmission media. The signal at the receiver is not identical to the signal at the transmitter.

- Attenuation
- Distortion
- Noise

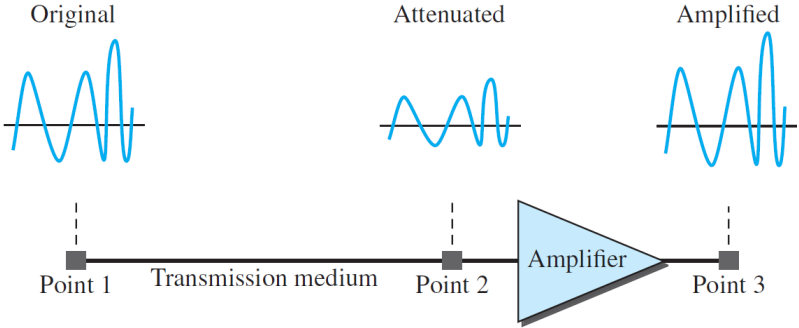


Attenuation

Definition

Attenuation means loss of energy as the signal travels through the medium.

- Energy converted to heat
- Signal amplitude decreases
- Amplifiers compensate for loss



Decibel (dB): Relative Attenuation/Gain

Power-Based Definition

$$\text{Power (dB)} = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

- P_1 = Initial power
- P_2 = Final power
- Negative $\rightarrow$ Attenuation
- Positive $\rightarrow$ Gain

Voltage-Based Definition

$$dB = 20 \log_{10} \left(\frac{V_2}{V_1} \right)$$

Problems: Attenuation, Gain and Characteristic Loss

Problem 1

Suppose a signal travels through a transmission medium and its power is reduced to one-half. Calculate attenuation.

Problem 2

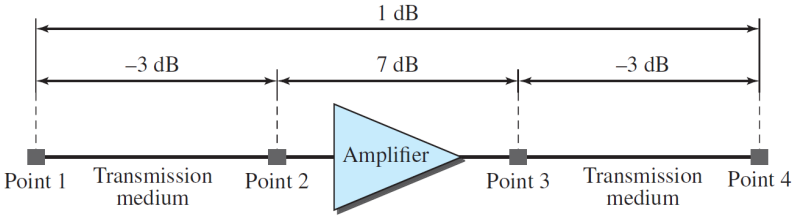
A signal travels through an amplifier, and its power is increased 10 times. Calculate gain.

Problem 3

The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km in mW?

Advantages of Using dB

- Converts multiplication into addition
- Simplifies cascade calculations
- Handles wide dynamic ranges



Received Signal Strength

Received Signal Strength (RSS)

RSS is a measurement of the **received signal power** at a receiver. It is typically expressed in **dBm**.

What is dBm?

- dBm = absolute power referenced to 1 mW
- 0 dBm = 1 mW, Negative dBm $\Rightarrow$ power $<$ 1 mW, Positive dBm $\Rightarrow$ power $>$ 1 mW

$$P(dBm) = 10\log_{10}\left(\frac{P(mW)}{1\,mW}\right)$$

When gain/loss is in dB and signal level is in dBm:

$$\text{Output (dBm)} = \text{Input (dBm)} + \text{Gain/Loss (dB)}$$

Receiver Sensitivity and RSSI

Receiver Sensitivity

Receiver Sensitivity is the **minimum received power level (in dBm)** at which a receiver can correctly detect and decode a signal with a specified performance (e.g., BER or PER).

Received Signal Strength Indicator (RSSI)

Device/vendor/operator-specific indicator of received power mapped to dBm, but often internally scaled, instead of an absolute value

Key Differences

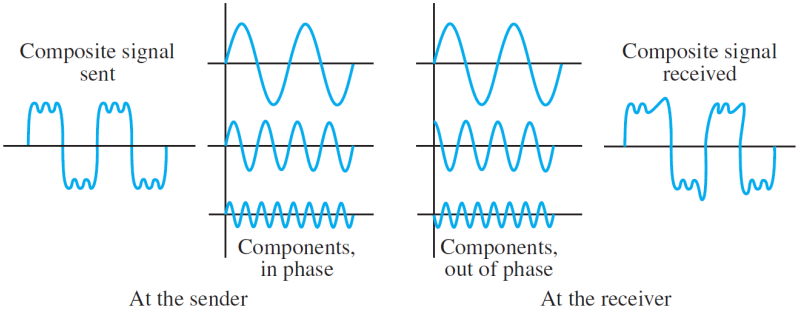
- **RSS:** Actual received power (absolute, in dBm)
- **Sensitivity:** Minimum required power for acceptable performance
- **RSSI:** Vendor-specific representation of received power

Distortion

Definition

Distortion means change in signal shape.

- Composite signals contain multiple frequencies
- Each frequency has different propagation delay
- Phase differences change signal shape



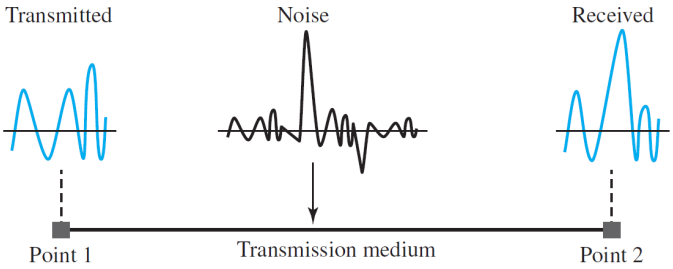
Noise

Definition

Noise is an unwanted signal added to the original signal.

Types of Noise:

- Thermal noise
- Induced noise
- Crosstalk
- Impulse noise

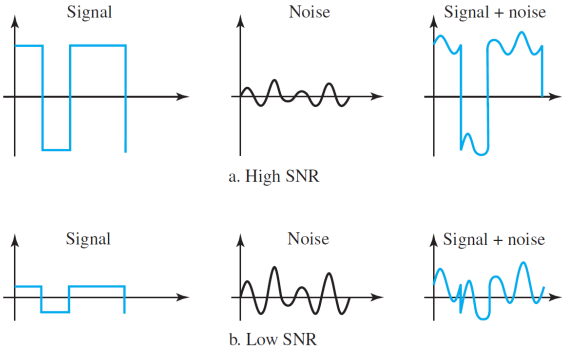


Signal-to-Noise Ratio (SNR)

Definition

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}}$$

$$SNR(dB) = 10\log_{10}\left(\frac{\text{Signal Power}}{\text{Noise Power}}\right)$$



Practical Perspective

Minimum SNR Requirements (Typical Values)

- BPSK/QPSK: $\approx$ 8–10 dB
- 16-QAM: $\approx$ 16–18 dB
- 64-QAM: $\approx$ 22–24 dB
- 256-QAM (5G/WiFi 6): $\approx$ 30 dB or higher

Common Causes of Impairments

- **5G:** Path loss at mmWave frequencies, blockage, beam misalignment
- **WiFi:** Co-channel interference, multipath fading, adjacent channel interference
- **Ethernet:** Cable attenuation, impedance mismatch, electromagnetic interference (EMI)

Summary

- Transmission impairment includes attenuation, distortion, and noise
- dB measures relative gain/loss
- dBm measures absolute power
- SNR determines signal quality

Reference

B. A. Forouzan, *Data Communications and Networking*, 5th ed. New York, NY, USA: McGraw-Hill, 2013.